

Assignment 1

One Walk, A Lifetime of Impact

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Visualization & Storytelling

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Introduction

Data is everywhere but without a story it means very little. Within the raw numbers, the percentages, the dots on a map, the peaks of a graph, the plateauing of a curve, the projections, the what if's and everything else that comes with the analytics and computations lives a human experience.

For this assignment, I wanted to create a story which humanizes the analysis and computations. Often, when knee-deep in the analysis and computations, it's easy to forget that these numbers translate to a person, a moment and a life.

There is no doubt that analysis and computations are of great significance, but they cannot stand alone. Remembering analytics and computations would not exist if it wasn't for the recording of a human's event/activity. Without a "subject" to study, there would be no data to analyse and/or compute. Most importantly just as the "subject" became a number in the study; we too can become a number (a statistic in the vast deep sea of computation/analytics) in someone else's study. We are no exception.

We all have family, friends, jobs, pets, school and bills. We can all relate on some level to the "subject" being studied. Our lives are all measurable in some way. My aim for this assignment is to unpack the data and tell a story that is not only informative, but also relatable and deeply human.

Topic Selection Rationale

Like many in this class, I had no clue what dataset I'll explore. Not because I lack interests but because my interests are not exactly "topical". As someone who considers themselves a bit of a news- connoisseur I knew I'd find inspiration in there. I can always count on the news reporters to bring the hot topics and bad news all within a fifteen-minute segment.

And like clockwork it came. Standing close enough to the intersection to see the yellow tape and covered body, the reporter announced, "...*there has been an accident on Bathurst and the pedestrian involved was dead. The driver remained on the scene*".

My heart immediately sank, as the reporter delivered the details in a *just-part-of-a-day's-work* tone. It hit me, "*that could have been me on either side of the incident*". It was something relatable but was it topical?

A few days later, Mayor Chow was back in the news defending speed cameras while Premier Ford continued pushing to have them removed from Ontario. Mayor Chow has used speed cameras as an instrumental tool in Toronto's Vision Zero Initiative.

This was it! I found a theme that was both topical and relatable. I believe I struck gold (or the assignment equivalent of gold- 15%). I knew this was the story for me to tell. I needed to find the *who, what, where, when, why* of the data while crafting a story using *the clear hook, body, climax and resolution*. With that I set out on recovering and exploring my dataset.

Dataset

For the scope of this assignment three sources of information were considered:

1. National Collision Database- There was an extraction utility and the raw data set
2. National Collision Database (Open Data): *Contains all the data for all collisions*
3. Toronto Open Data- *Motor Vehicle Collisions involving Killed or Seriously Injured Persons.*

For more information on these please see the “Citations” portion of this document

Audience

1. Policy Makers- City of Toronto decision makers and policy staff
2. Law Enforcement: Traffic enforcement unit that enforces speeding

Methods

Phase 1 (Cleaning the Data):

For this iteration my first step was to understand what data I had in my possession. I began “wrangling” (cleaning) the data and uncovered the: *Who*, *What*, *Where*, *When* and *Why* of each collision. I refer to these categories as my “buckets”.

Phase 2 (Organizing the Data):

In this phase, I used Miro to form buckets (logical groupings aka themes) and sorted the data elements into their respective buckets. For visualization, each data element is beneath their corresponding buckets/themes. This allowed me to clearly identify which database elements corresponded to which bucket. Below is a sample of what that process looked like.



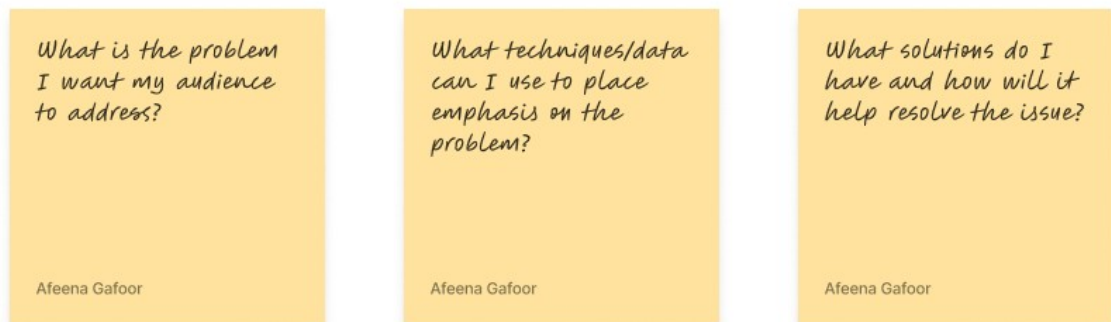
Phase 3 (Understanding the data):

For each bucket, I analysed the data it contained to gain a deeper understanding of what it was indicating.

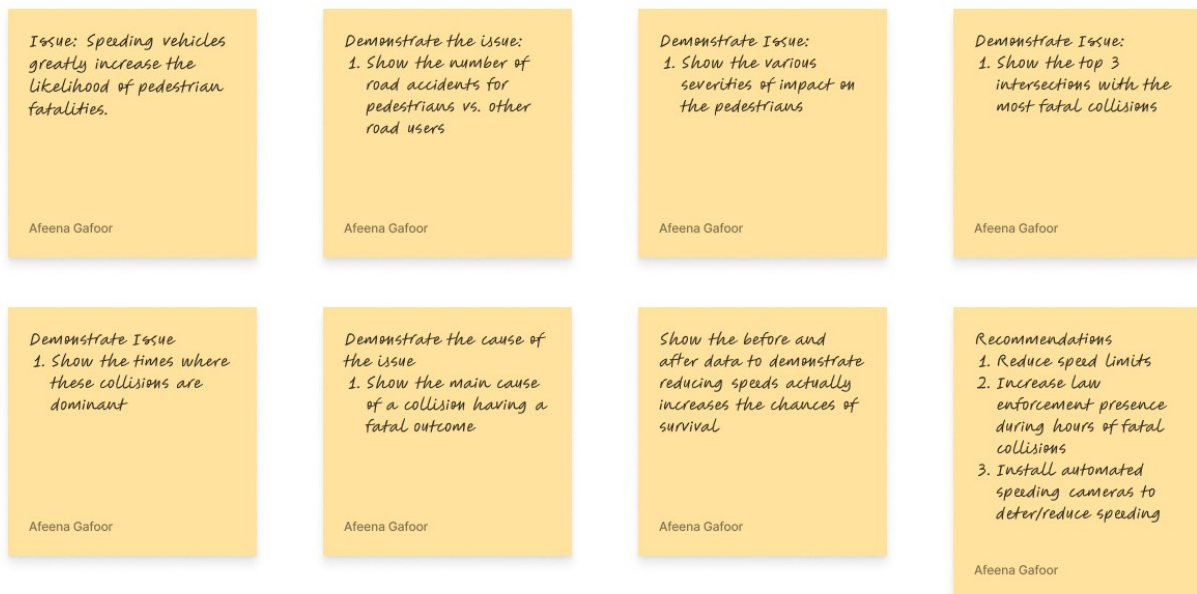
Phase 4 (Storyboarding):

For this task, I chose Figma because it helps me to organize and reorder my ideas as I go.

- **Iteration 1:** I identified the framework I wanted to build around and established the foundation for how the story and visuals would take shape.

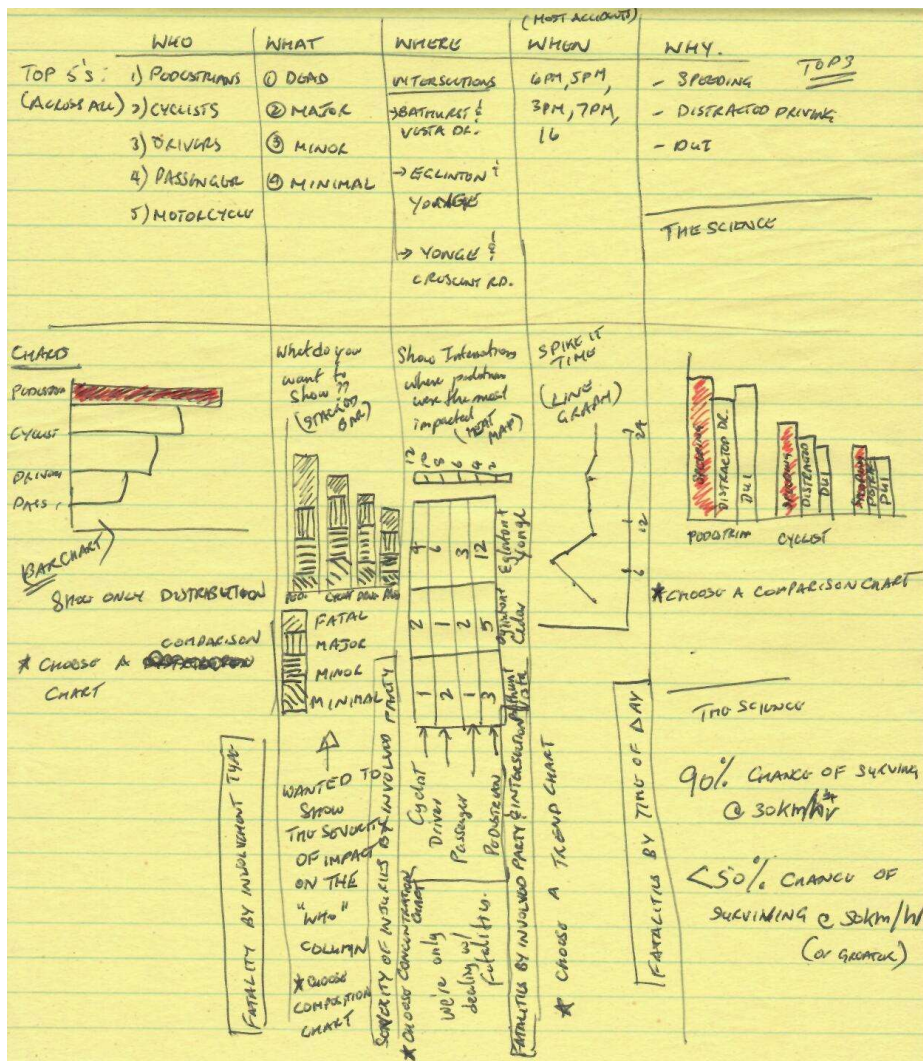


- **Iteration N:** After several iterations I was able to organize the story into a clear logical flow. It progresses from Identifying the issue, to demonstrating the problem, then presenting it's resolution and impact and finally closing with recommendations.



Phase 5 (Graph Selection for Story)

- **Iteration 1:** Based on the insights I wanted to demonstrate, I selected the appropriate chart (see Appendix A).
 - To keep the audience focused on the story rather than the number crunching I chose to highlight only the top 3 (or top 5) statistics (Chart 5)
 - Colours: Used red to reflect where I wanted their eyes to go. I chose red since we are dealing with fatalities (Chart 1, Chart 5)



- **Iteration N-** Revamping the charts to align with the storyboard objectives

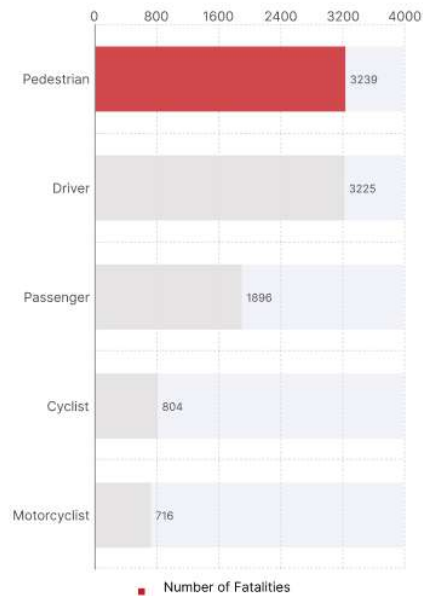
My aim was to design charts rich in data yet not overwhelming the audience with too much content. While exploring options that best supported the story's narrative, I recognised that the placeholder charts were severely lacking in what I was trying to convey. To create a seamless and cohesive flow between narrative frames, I decided to mock up the charts in Figma using the *iCharts generate* plugin.

Below is a mapping of each storyboard component to a chart, followed by my rationale behind my chart layout

Chart 1 (horizontal bar)

Demonstrate the issue:
1. Show the number of road accidents for pedestrians vs. other road users

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Source: Open Data- City of Toronto, Motor Vehicle collisions involving killed or serious injured persons

- Retained dark red to represent fatal outcomes
- Went for the top 5 to keep my audience focused
- Faded all other categories into the background to keep the focus on the pedestrians
- Included a legend for colour scheme understanding
- Placed exact figures beside the bars to provide precise numbers rather than estimates of fatalities
- Included a source of the statistics, intentionally styled in small semi-transparent text to avoid distracting the audience
- *Plugin limitations:* Grid lines could not be removed. This would have been possible if the charts were created using Plotly. Additionally, I would have positioned the x-axis at the bottom.
- *Note: the numbers on these charts are arbitrary as the focus is on design rather than analytics*

Chart 2 (composite bar)

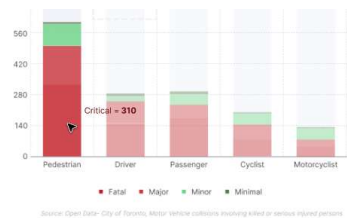
Demonstrate Issue:
1. Show the various severities of impact on the pedestrians

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- Same as above
- Used two distinct shades of red to differentiate between major and critical
- Applied two shades of green for the minimal and minor segments. Green indicating “good-to-go”

Chart 2.1-interactive



- Interactive Effect: Upon hover, the count of the segment would appear

Chart 3 (heatmap)

Demonstrate Issue:
1. Show the top 3 intersections with the most fatal collisions

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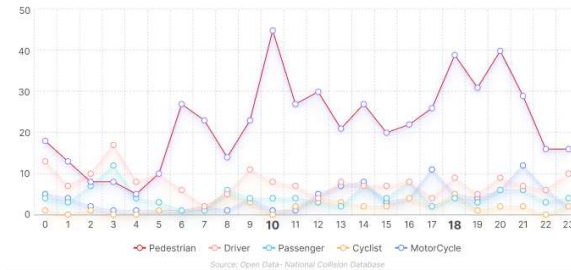


- This chart was handmade
- Used distinct shades of red which increases in intensity as the numbers of fatalities increase
- Included a source of the statistics, intentionally styled in small semi-transparent text to avoid distracting the audience
- *Note: the numbers on these charts are arbitrary as the focus is on design rather than analytics*

Chart 4 (line chart)

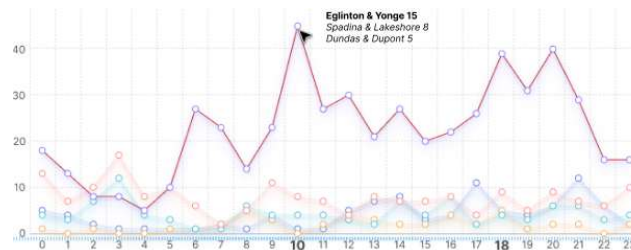
*Demonstrate Issue
1. Show the times where
these collisions are
dominant*

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- Made the line for the pedestrians red for emphasis
- Faded all other categories into the background to keep the focus on the pedestrians
- Included a legend for colour scheme understanding
- Intentionally bolded 10 and 18 to tie it into my story
- Included a source of the statistics, intentionally styled in small semi-transparent text to avoid distracting the audience
- Included the grid lines for ease of pinpointing peaks.
- *Note: the numbers on these charts are arbitrary as the focus is on design rather than analytics*

Chart 4.1-
interactive



- Interactive Effect: Upon hover, the top 3 intersections that contributed to the death toll are highlighted, tying the visualisation back to the heatmap above

Chart 5 (composite bar)

Demonstrate the cause of the issue
1. Show the main cause of a collision having a fatal outcome

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- Retained dark red to represent fatal outcomes
- Faded all other categories into the background to keep the focus on the pedestrians
- Included a legend for colour scheme understanding
- Placed exact figures beside the bars to provide precise numbers rather than estimates of fatalities
- Included a source of the statistics, intentionally styled in small semi-transparent text to avoid distracting the audience
- Plugin limitations: Gridlines could not be removed. This would have been possible if the charts were created using Plotly.
- *Note: the numbers on these charts are arbitrary as the focus is on design rather than analytics*

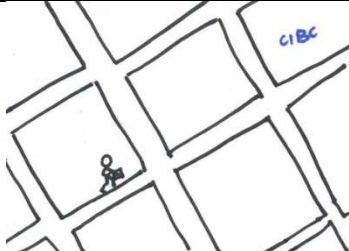
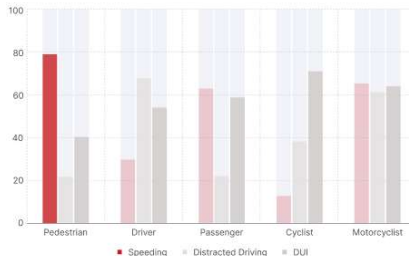
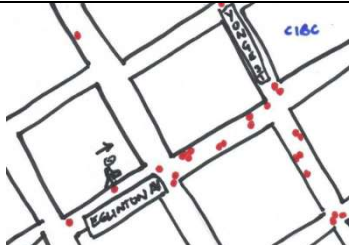
Phase 6 (Pulling it together):

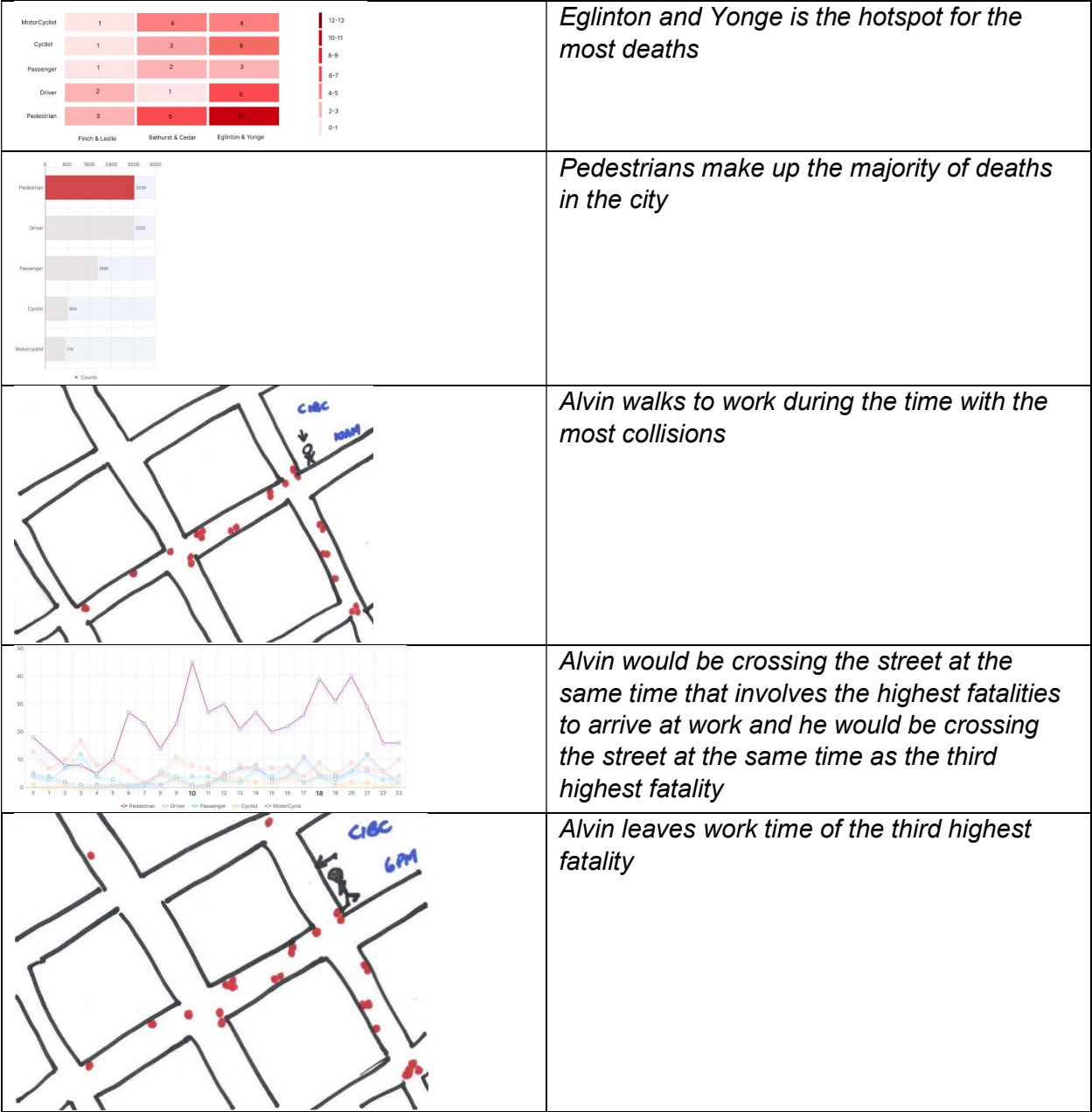
Now that I know the key information, I can begin shaping a narrative that draws the audience in. I also have a clear sense of who my audience is. They are- fathers, mothers and they can all relate to walking on the street.

It is now time for me to frame the story of my protagonist within the following framework: a *compelling hook, a strong body, climax and thoughtful resolution*- just as any good storyteller would.

In this iteration I resorted to creating small stick figure sketches of my protagonist and how he interacts with his environment. I scanned those sketches and used Figma which allows for the flexibility to reorder the drawings and graphs pulling everything together.

- **Iteration 1:** This is what I had initially but after looking at it still did not invoke any compassion. It was just empty emotionless.

	<i>This is Alvin, his wife Tiffany and his 10 month old daughter Lulu</i>																								
	<i>Everyday Alvin walks to work</i>																								
 <table><tr><th>Vehicle Type</th><th>Speeding (%)</th><th>Distracted Driving (%)</th><th>DUI (%)</th></tr><tr><td>Pedestrian</td><td>80</td><td>40</td><td>40</td></tr><tr><td>Driver</td><td>30</td><td>65</td><td>55</td></tr><tr><td>Passenger</td><td>60</td><td>20</td><td>55</td></tr><tr><td>Cyclist</td><td>10</td><td>35</td><td>70</td></tr><tr><td>Motorcyclist</td><td>65</td><td>60</td><td>65</td></tr></table>	Vehicle Type	Speeding (%)	Distracted Driving (%)	DUI (%)	Pedestrian	80	40	40	Driver	30	65	55	Passenger	60	20	55	Cyclist	10	35	70	Motorcyclist	65	60	65	<i>Pedestrians have proven to show the fatal consequences when involved in a collision. My charts at this time had no data source.</i>
Vehicle Type	Speeding (%)	Distracted Driving (%)	DUI (%)																						
Pedestrian	80	40	40																						
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Cyclist	10	35	70																						
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	<i>Alvin's work is located at the intersection that has the most deaths</i>																								



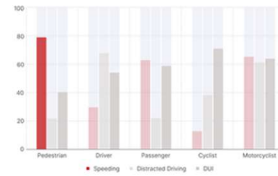
THE FULL STORY (ITERATION 1):



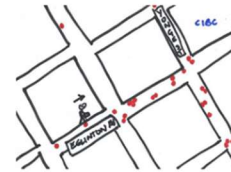
This is Alvin, his wife Tiffany and his 10 month old daughter Cindy



Everyday Alvin walks to work



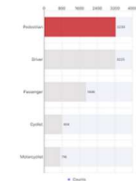
Pedestrians have proven to show the fatal consequences when involved in a collision



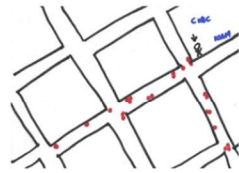
Alvin's work is located at the intersection that has the most deaths

Motorcyclist	1	4	4
Cyclist	1	3	4
Passenger	1	2	3
Driver	2	1	4
Pedestrian	3	5	6

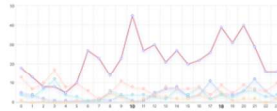
Eglinton and Yonge is the hotspot for the most deaths



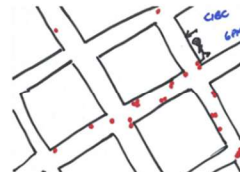
Pedestrians make up the majority of deaths in the city



Alvin walks to work during the time with the most collisions



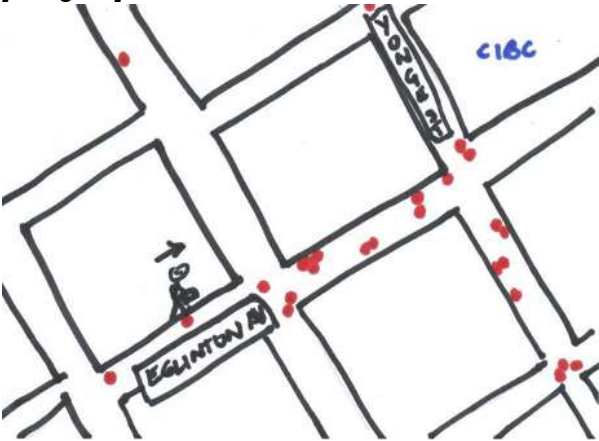
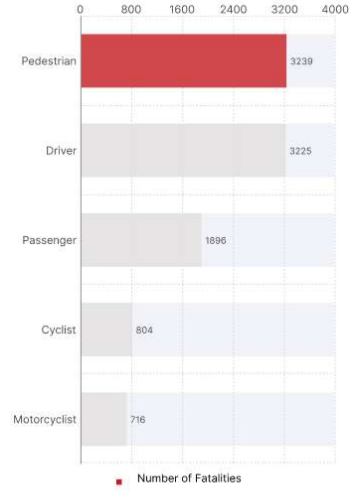
Alvin would be crossing the street at the same time that involves the highest fatalities to arrive at work and he would be crossing the street at the same time as the third highest fatality



Alvin leaves work time of the third highest fatality

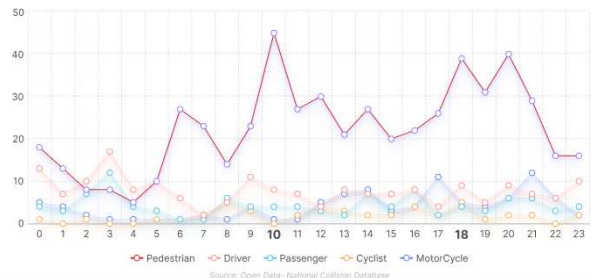
- Iteration N:** After completing iteration 1, I realised more work was needed to highlight the human element. While it can be challenging to convey emotions through stick figures, I reworked my drawings to help the audience empathize more easily. I also refined my narration to make it less “matter of fact” and made it into more of a story. Additionally, I added scrollytelling techniques and chart interactions to enhance the overall impact on the audience.

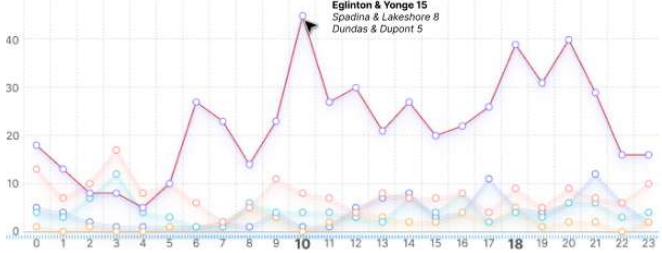
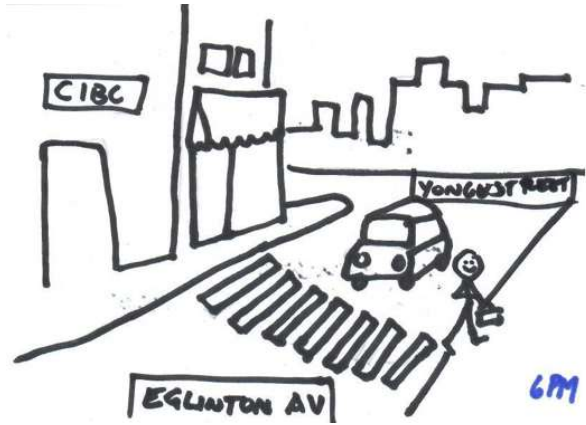
<i>[The story starts with this image]</i> 	<i>[Scroll]</i> <i>“Meet Alvin, a hardworking newlywed. His wife Jane and their baby Lulu are at the heart of his world”</i>	<i>[Next Scroll]</i> <i>Introduce Image 2</i>	<i>[Purpose]</i> <i>Establishes the hook and pulls in the family dynamic. Added a small baby to amplify vulnerability.</i>
<i>[Image 2]</i> 	<i>[Scroll]</i> <i>“Every morning Alvin walks 17 minutes to work through the busy streets of Eglinton Avenue. On his way, Alvin FaceTimes Jane and Baby Lulu- a quick call because he already misses them and they miss him even more”</i>	<i>[Next Scroll]</i> <i>Introduce Image 3</i>	<i>[Purpose]</i> <i>Deepens the loving relationship between Alvin and his family. Highlighting that it is a reciprocal relationship.</i>

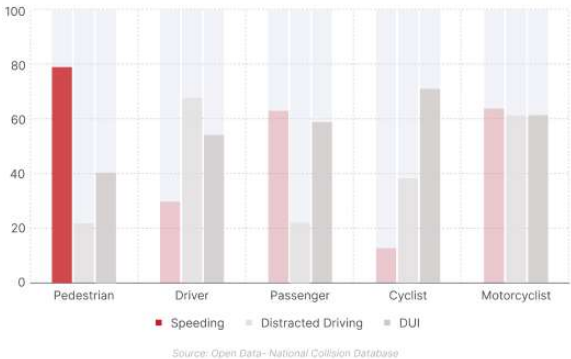
<i>[Image 3]</i> 	<i>[Scroll]</i> <i>“These aren’t just numbers. Each red dot marks a pedestrian whose life was snuffed out on Eglinton Avenue</i> <i>For families like Jane and baby Lulu, every step Alvin takes carries a risk shaped by our city’s policies”</i>	<i>[Next Scroll]</i> <i>Introduce Graph 1</i>	<i>[Purpose]</i> <i>Transitioning from family intimacy to the road risks. Grounding the story in Alvin’s own street.</i>
<i>[Graph 1]</i>  Source: Open Data- City of Toronto, Motor Vehicle collisions involving killed or serious injured persons	<i>[Scroll]</i> <i>“What happens on Eglinton is not unique. Across Toronto, pedestrians consistently face the highest risk of being killed”</i>	<i>[Next Scroll]</i> <i>Introduce Graph 2</i>	<i>[Purpose]</i> <i>Placement of this slide is to introduce the data after the emotional setup. Shows data contrast clearly. Hopefully the audience made the connection between Alvin and the numbers</i>
<i>[Graph 2]</i>	<i>[Scroll]</i>	<i>[Next Scroll]</i>	<i>[Purpose]</i>

 Source: Open Data- City of Toronto, Motor Vehicle collisions involving killed or seriously injured persons	<i>“For non-pedestrians, these collisions end in less fatal outcomes. But for pedestrians the outcome is far more severe. It can be fatal or even have life changing impacts”</i>	<i>Introduce Graph 2.1- Interactive</i>	<i>Placed this chart here to show that pedestrians not only experience fatalities but also suffer the most major injuries as well.</i>
[Graph 2.1]  Source: Open Data- City of Toronto, Motor Vehicle collisions involving killed or seriously injured persons	[Scroll] <i>“Let’s take a closer look. When we hover over each group we see the raw numbers”</i>	[Next Scroll] <i>Introduce Image 4</i>	[Callouts] <i>Notice how pedestrians have their numbers stacked heavily towards the most severe outcome</i> <i>Interactions shift the audience from observer to participant.</i>

<i>[Image 4]</i> 	<i>[Scroll]</i> <i>“Now imagine Alvin crossing right here at Yonge & Eglinton, one of the busiest intersections in the city”</i>	<i>[Next Scroll]</i> <i>Introduce Graph 3- Heatmap</i>	<i>[Purpose]</i> <i>A zoom in moment. Placing Alvin at a high-risk corridor. Sets the scene for my next slide</i>
<i>[Graph 3]</i>  Source: Open Data - National Collision Database	<i>[Scroll]</i> <i>“Yonge and Eglinton isn’t just busy it’s where the most pedestrians have lost their life than any other intersection in the City of Toronto. It’s the same corner that Alvin crosses twice a day at 10 AM and 6PM”</i>	<i>[Next Scroll]</i> <i>Introduce Image 5</i>	<i>[Purpose]</i> <i>Data punch! Zooming out from the Yonge & Eglinton (most dangerous) location and showing this happens on other Toronto streets</i>
<i>[Image 5]</i>	<i>[Scroll]</i> <i>“Every morning at 10AM Alvin crosses here on his way to work”</i>	<i>[Next Scroll]</i> <i>Introduce Graph 4</i>	<i>[Purpose]</i> <i>Setting the scene for my next slide. Where I look at</i>

			the fatalities at specific times [Lead in] ...but 10AM is the time when most pedestrians are killed
[Graph 4] 	[Scroll] “When we look at fatal collisions across the day, one trend stands out: pedestrians are impacted the most above all other road users at 10 AM and 6 PM”	[Next Scroll] Introduce Graph 4.1	[Purpose] Setup to show Alvin’s routine overlaps with the deadliest hours [Callout] Draw attention to the 10 AM hour and 6PM which has elevated fatalities
[Graph 4.1]	[Scroll] “This graph isn’t just numbers. We can click on any point to reveal the top three	[Next Scroll] Introduce Image 6	[Purpose] Ties the intersection back to the time of day and hopefully Alvin.

	<i>intersections contributing to these fatalities</i> <i>Again and again, Yonge and Eglinton appears- the very corner Alvin crosses each day”</i>		<i>Makes the audience aware of the intersections that are the riskiest</i>
<i>[Image 6]</i> 	<i>[Scroll]</i> <i>“At 6PM Alvin leaves work and begins his walk home.</i> <i>It should be the safest moment of the day heading back home to Jane and baby Lulu.</i> <i>But 6PM is also one of the deadliest hours for pedestrians with fatalities peaking”</i>	<i>[Next Scroll]</i> <i>Introduce Image 7</i>	<i>[Purpose]</i> <i>Ties Alvin back to the data on the line chart. Shows the audience that Alvin again has to take the route, not just once, but twice a day.</i>
<i>[Image7]</i> 	<i>[Scroll]</i> <i>“The dangers Alvin faces are not random. Across Toronto, there are 3 major causes of road fatalities: Speeding, Distracted Driving and Driving Under the Influence”</i>	<i>[Next Scroll]</i> <i>Introduce Graph 5</i>	<i>[Purpose]</i> <i>This slide is to show the main cause of the fatalities. The “why” element of the story.</i>
<i>[Graph 5]</i>	<i>[Scroll]</i>	<i>[Next Scroll]</i>	<i>[Purpose]</i>

 Source: Open Data - National Collision Database	<i>“When we break the fatalities down by cause, there is one clear fact: pedestrians die the most.</i> <i>While speeding, distracted driving and DUI put everyone at risk, they are most deadly for people like Alvin-pedestrians”</i>	<i>Introduce Image 8</i>	<i>Data kick to show that speed is the main cause of death to pedestrians. Hopefully invoking some feelings on the audience to take action.</i>
<i>[Image 8]</i> 50 KM/HR	<i>[Scroll]</i> <i>“At the city speeds, pedestrians have a 50% chance of surviving. For every pedestrian struck at 50km/hr half will not make it”</i>	<i>[Next Scroll]</i> <i>Introduce Image 9</i>	<i>[Purpose]</i> <i>Shows the mathematics of how speed is directly related to cause of deaths. Alvin’s life has been built and risks identified now for the final blow to the audience</i>
<i>[Image 9]</i>	<i>[Scroll]</i> <i>“In this story Alvin is one of them. A simple walk to work ends in tragedy”</i>	<i>[Next Scroll]</i> <i>Introduce Image 10</i>	<i>[Purpose]</i> <i>To show the worst-case scenario</i>

			
<i>[Image 10]</i> 	<i>[Scroll]</i> <i>“While Alvin is gone the real burden falls on the ones left behind. Jane loses her husband and Lulu never got to know her father”</i>	<i>[Next Scroll]</i> <i>“....The other alternative”</i>	<i>[Purpose]</i> <i>I placed the focus back on his family. They now navigate life without Alvin. Notice, I showed the baby smiling to still depict innocence while the mother is devastated.</i>
<i>[Image 11]</i> 30 KM/HR	<i>[Scroll]</i> <i>“But lowering the speed changes everything. At 30KM/HR 9 out of 10 pedestrians survive”</i>	<i>[Next Scroll]</i> <i>Introduce Image 12</i>	<i>[Purpose]</i> <i>For contrast I wanted to show that there is another alternative to Alvin dying. Trying to make the policy makers feel a</i>

			<i>little responsible for their choices</i>
[Image 12] 	[Scroll] <i>“At slower speeds a collision doesn’t have to mean death. Alvin is shaken but he survives”</i>	[Next Scroll] <i>Introduce Image 13</i>	[Purpose] <i>To show Alvin survives with no serious injuries</i>
[Image 13] 	[Scroll] <i>“Instead of Jane opening the door to tragedy, she opens it to Alvin- alive, smiling, carrying a cake and a balloon for his family.</i> <i>Because with safer speeds, Lulu keeps her father and Jane keeps her husband”</i>		[Purpose] <i>To show the best-case scenario and how policy makers can directly impact the lives of families.</i>

THE FULL STORY (ITERATION N):



"Must Alvin, a hardworking lawyer. His wife Jane and baby Lily are at the heart of the world"



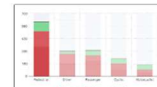
Every morning Alvin walks 17 minutes to work through the busy streets of Eglinton Avenue. On his way, Alvin FaceTimes Jane and Baby Lily in a quiet call because he already misses them and they miss him even more



These aren't just numbers. Each red dot marks a pedestrian whose life was snuffed out on Eglinton Avenue.
For families like Jane and Baby Lily, every step Alvin takes comes with a shadow by our daily collisions



These aren't just numbers. Each red dot marks a pedestrian whose life was snuffed out on Eglinton Avenue"



For non-pedestrians, these collisions and in less time exposure. But for pedestrians the outcome is far more serious. It can be fatal or even have life-changing impacts



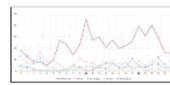
Now imagine Alvin crossing right here at Yonge & Eglinton, one of the busiest intersections in the city



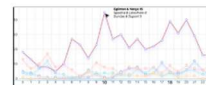
Yonge and Eglinton isn't just busy. It's where the most pedestrian deaths occur. The four other intersections in the city of Toronto. It's the same corner that Alvin crosses every day at 10 AM



Every morning at 10 AM Alvin crosses here on his way to work



10 AM records the highest fatality rates. 6 PM records the third highest fatality rates



This graph isn't just numbers. We can click on any point to reveal up to 100 more intersections contributing to these fatalities.

Again and again, Yonge and Eglinton appears - the very corner Alvin crosses each day



At 6 PM Alvin leaves work and begins his walk home. It should be the safest moment of the day heading back home to see his baby Lily.

But 6 PM is also one of the deadliest hours for pedestrians with families passing



The dangers Alvin faces are not random. Across Toronto, there are 3 major causes of road fatalities: Speeding, Distraction Driving and Drinking Under the Influence



When we break the fatalities down by cause, there is one clear fact: pedestrians die the most

While speeding, distraction driving and DUI put everyone at risk, they are most deadly for people like Alvin

50 KM/HR



In this story Alvin is one of them. A simple walk to work ends in tragedy



While Alvin is going the safe direction, falls on the crosswalk corner. Jane loses her husband and Lily never got to know her father

30 KM/HR



At slower speeds a collision doesn't have to mean death. Alvin is injured but he survives



Instead of Jane covering the cost of tragedy, she reports it to Alvin's insurer, getting a cash award to help her family. Rebecca with safer speeds, Lily keeps her father and Jane keeps her husband

But lowering the speed changes everything. At 30 KM/HR 9 out of 10 pedestrians survive

One Walk, A Lifetime of Impact.

Appendix A

Objective	Chart	Purpose
Comparison	Bar chart (horizontal/vertical)	Category comparisons
	clustered bar chart	Compare categories across subcategories
	Line chart	Show change between points
Distribution	Histogram	Show frequency of distribution
	Box plot	Show spread, median, outlier
	scatterplot	Relationship between two variables
Trend	Line chart	Showing data over time
	Area chart	Used over cumulative values
Highlight outliers	Scatterplots with callout	Show unusual points
	Single number / big bold text	

Citations

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